Algebraga kirish

1-QISM

Qiziqtiramiz

1-SAVOL · TAKSI

Shahar taksisiga o'tirishning o'zi 6 000 so'm. Har bir kilometr uchun yana 2 500 so'm to'lanadi.

4 km yo'l necha so'm turadi? 10 km-chi? Yo'l necha kilometr bo'lishini hali bilmasangiz, narxni qanday yozib qo'yasiz?

2-SAVOL · GUGURT CHO'PLARI



1-shakl
3 ta cho'p

2-shakl
5 ta cho'p

3-shakl
7 ta cho'p

Gugurt cho'plaridan qatorasiga uchburchaklar terildi. 10 ta uchburchak uchun nechta cho'p kerak? Rasmini chizmasdan topa olasizmi?



Kilometrlar soni har safar boshqacha. Har safar qaytadan hisoblamay, hamma holatga yaraydigan bitta yozuv topsak-chi?

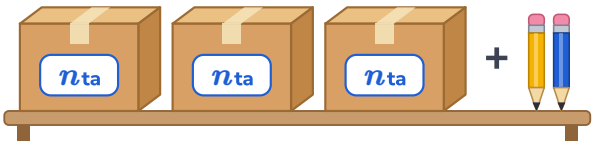
ESLAB OLAYLIK

- Amallar tartibi: avval qavs, keyin daraja, keyin ko'paytirish va bo'lish, oxirida qo'shish va ayirish: $6\,000 + 2\,500 \cdot 4 = 16\,000$.
- Daraja: $3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$ — 3 daraja asosi, 4 daraja ko'rsatkichi.
- Ishoralar: $5 \cdot (-3) = -15$, $(-3)^2 = 9$, $(-2)^3 = -8$.

2-QISM

Sinab ko'ramiz

1-TOPSHIRIQ · JAVONDAGI QUTILAR



Uchta yopiq qutida bir xil sondagi qalam bor — yorlig'ida faqat «n ta». Yonida yana 2 ta qalam.

a) Qutilarni ochmasdan jami qalamlar sonini qanday yozasiz? b) Har qutida 12 ta qalam bo'lsa, jami nechta?

2-TOPSHIRIQ · JADVALNI TO'LDIRING

n	1	2	3	4	5
$n + n + n$	3	?	?	?	?
$3 \cdot n$	3	?	?	?	?
$n \cdot n$	1	?	?	?	?

Qaysi ikki qator hamma ustunda bir xil chiqdi? Uchinchi qator ular bilan qayerda teng bo'ldi?

3-TOPSHIRIQ · CHO'PLAR QONUNIYATI

2-savoldagi uchburchaklarga qarang: 4 va 5 ta uchburchak uchun nechta cho'p kerak? Har yangi uchburchak uchun nechta cho'p qo'shiladi? Qaysi cho'p hamma shaklda «ortiqcha» bo'lib qoladi?

3-QISM

Tushunamiz

TA'RIF · O'ZGARUVCHI VA ALGEBRAIK IFODA

Arifmetikada biz aniq qiymatga ega sonlar ustida amallar bajardik. Algebrada sonlarni ifodalash uchun $a, b, c, \ldots$ harflaridan ham foydalanamiz; harflar har qanday sonli qiymatlarni anglatishi mumkin.

$6\,000 + 2\,500 \cdot n$ ifodada kilometrlar sonini ko'rsatish uchun n harfini qo'lladik. n — **o'zgaruvchi**, $6\,000 + 2\,500 \cdot n$ esa **algebraik ifoda** deb ataladi.

Algebraik ifodalar amal belgilari va qavslar bilan birlashtirilgan sonlar va o'zgaruvchilardan tuziladi.

1-MISOL · 1-SAVOL

4 km: $6\,000 + 2\,500 \cdot 4 = 16\,000$ so'm.
10 km: $6\,000 + 2\,500 \cdot 10 = 31\,000$ so'm.
n km: $(6\,000 + 2\,500n)$ so'm — bitta yozuv hamma yo'lga yaraydi.

QOIDA · KO'PAYTIRISH BELGISI

Harf bilan harf, son bilan harf orasidagi ko'paytirish belgisi — «×» yoki «·» nuqta ko'pincha tushirib qoldiriladi.

$a \cdot b = ab$, $3 \cdot y = 3y$, $1 \cdot y = y \cdot 1 = y$, $8 \cdot ab^2 = 8ab^2$. Son harfdan oldin yoziladi: $y \cdot 5 = 5y$.

Bo'lish kasr ko'rinishida yoziladi: $x : y = \dfrac{x}{y}$, bu yerda $y \neq 0$.

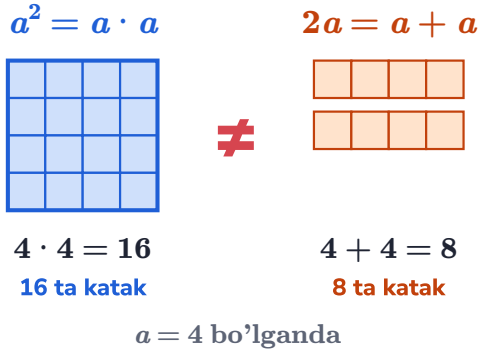
Amal	Aytilishi	Yozilishi
Qo'shish	p plyus q; p va q ning yig'indisi	$p + q$
Ayirish	p minus q; p dan q ni ayirish	$p - q$
Ko'paytirish	p va q ning ko'paytmasi; p ni ikkilangani	pq; $2p$
Bo'lish	p ning q ga bo'linmasi, $q \neq 0$	$\dfrac{p}{q}$
Qarama-qarshi son	p ning qarama-qarshisi	$-p$

QOIDA · DARAJA

Algebrada daraja belgisi arifmetikadagidek qo'llanadi:

$a \cdot a = a^2$ — «a kvadrat» yoki «a ning kvadrati»;
$b \cdot b \cdot b = b^3$ — «b kub» yoki «b ning kubi»;
$c \cdot c \cdot c \cdot c = c^4$ — «c ning to'rtinchi darajasi».

E'tibor bering: $a^2 \neq a + a$, demak $a^2 \neq 2a$; $b^3 \neq 3b$; $c^4 \neq 4c$.



2-MISOL · JUMLADAN IFODAGA

a «p ning kvadratini q ga ko'paytirib, natijadan r ni ayirish»: kvadrat p^2, ko'paytma p^2q; natija $p^2q - r$.

b «m va n ning yig'indisini ularning ko'paytmasiga bo'lish»: yig'indi $m + n$ butunligicha bo'linadi — $\dfrac{m+n}{mn}$.

Ifodaning qiymatini hisoblash

TA'RIF · O'RNIGA QO'YISH

Algebraik ifodadagi har bir o'zgaruvchining son qiymatini bilsak, ifodaning qiymatini hisoblay olamiz.

Algebraik ifodaning qiymatini topish uchun har bir o'zgaruvchini uning son qiymati bilan almashtirish jarayoni o'rniga qo'yish deyiladi.

Usul. Manfiy son va kasrni qavs ichida qo'yamiz; tushirib qoldirilgan ko'paytirish belgisini qaytaramiz. Keyin amallar tartibi bilan hisoblaymiz.

3-MISOL · IKKI O'ZGARUVCHI

$x = -3$ $y = 2$

$$5 - 2xy$$

$$5 - 2 \cdot (-3) \cdot 2$$

$$= 5 + 12$$

$$= 17$$

4-MISOL · DARAJA BILAN

$x = -3$ $y = 2$

$$xy^2 - x^2$$

$$(-3) \cdot 2^2 - (-3)^2$$

$$= -3 \cdot 4 - 9$$

$$= -21$$

$(-3)^2 = 9$ — qavs bo'lmasa, $-3^2 = -9$ bo'lib, javob xato chiqadi.

USUL · TENG KUCHLI IFODALAR

Ikki ifodaga o'zgaruvchining turli qiymatlarini qo'yib, qiymatlarini taqqoslaymiz. O'zgaruvchining **barcha** qiymatlarida qiymatlari teng bo'lsa — ifodalar **teng kuchli**. Birorta qiymatda farq qilsa — teng kuchli emas.

2-topshiriq: $n + n + n$ va $3n$ — teng kuchli. $3n$ va n^2 esa $n = 0$ va $n = 3$ da bir xil qiymatga ega bo'lsa ham, $n = 1$ da $3 \neq 1$ — teng kuchli emas.

Formulalar

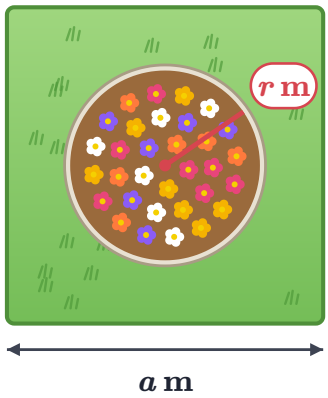
TA'RIF · FORMULA

Formula — ikki yoki undan ortiq o'zgaruvchini bir-biriga bog'lovchi tenglik.

To'g'ri to'rtburchakning eni a, bo'yi b, perimetri P bo'lsa: $P = 2(a + b)$ — formula; $2(a + b)$ — algebraik ifoda.

Javobni π orqali topishda π ning kalkulyator yordamida yoki yaxlitlangan 3,142 yoki $\frac{22}{7}$ qiymatlarini o'rniga qo'yib hisoblamaymiz. Yakuniy javob π belgi bilan yoziladi.

5-MISOL · MAYSAZOR



Kvadrat maydon tomoni a m, o'rtasida radiusi r m gulzor. Maysa yuzi: $S = a^2 - \pi r^2$.

$a = 10, r = 3$: $S = 10^2 - \pi \cdot 3^2 = (100 - 9\pi)$ m².

6-MISOL · 2-SAVOL

n ta uchburchak uchun cho'plar soni: $N = 2n + 1$.

$n = 10$: $N = 2 \cdot 10 + 1 = 21$ ta cho'p.

4-QISM

Qo'llaymiz

1-MISOL · QISQA YOZING

a) $x \cdot 5 \cdot y \cdot 3$: sonlarni ko'paytirib oldinga yozamiz $5 \cdot 3 = 15$, harflarni yonma-yon yozamiz: $15xy$.

b) $k \cdot 3 \cdot k \cdot k$: son — 3, $k \cdot k \cdot k = k^3$: $3k^3$.

c) $2m : n \cdot k$: chapdan o'ngga — avval $\frac{2m}{n}$, keyin k ga ko'paytiramiz (k suratga yoziladi): $\frac{2mk}{n}$.

d) $6p : 2q - r$: avval bo'lish: $\frac{6p}{2q} = \frac{3p}{q}$; natija $\frac{3p}{q} - r$ (r bo'linmaydi).

2-MISOL · LIFT

Lift ko'pi bilan 650 kg ko'taradi. Unga har biri m kg bo'lgan 4 ta quti va d kg li aravacha kiritildi. Lift yana qancha yuk ko'tara oladi?

1) Yuk: $4m + d$ kg. Qolgan imkoniyat: $650 - (4m + d)$ kg.

2) $m = 85$, $d = 40$ bo'lsa: $650 - (340 + 40) = 270$ kg.

3-MISOL · IKKI XIL HARF

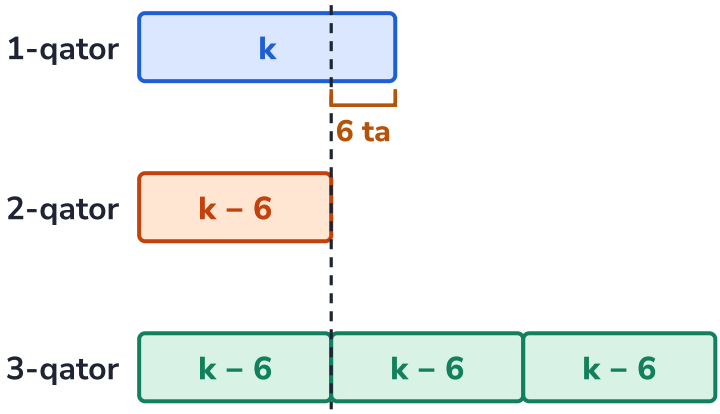


$$2m + k + 3$$

Ikki qopda m tadan, qutida k ta olma bor. Qop va qutidagi son har xil bo'lishi mumkin — shuning uchun harflar ham har xil.

4-MISOL · KO'CHAT QATORLARI

1-qatorga k ta ko'chat ekildi, 2-qatorga undan 6 ta kam, 3-qatorga esa 2-qatordagidan 3 marta ko'p.



2-qator: $k-6$. 3-qator: $3(k-6)$ — ayirma butunligicha 3 ga ko'paytiriladi, shuning uchun qavs kerak.

5-MISOL · QARAMA-QARSHI VA ILDIZ

a) «k ning qarama-qarshisiga 5 ni qo'shib, natijani m ning kvadratiga bo'lish»: $\dfrac{-k+5}{m^2}$.

b) «x ning kvadrat ildizini 4 ga ko'paytirib, natijadan y ni ayirish»: $4\sqrt{x}-y$.

6-MISOL · FORMULADAN TOPISH

a) $y=\dfrac{p^2-q}{p+q}$, $p=-4$, $q=2$: $y=\dfrac{16-2}{-2}=-7$.

b) $\dfrac{1}{x}=\dfrac{1}{a}+\dfrac{1}{b}$, $a=3$, $b=6$: $\dfrac{1}{x}=\dfrac{2}{6}+\dfrac{1}{6}=\dfrac{1}{2}$, demak $x=2$.

7-MISOL · KASR QIYMAT

$a=\dfrac{1}{3}$, $b=6$, $c=-4$ bo'lsa: a) $9a^2+bc$; b) $\dfrac{c-ab}{a+c}$.

a) $9\cdot\left(\dfrac{1}{3}\right)^2+6\cdot(-4)=9\cdot\dfrac{1}{9}-24=1-24=-23$.

b) Surat: $-4-\dfrac{1}{3}\cdot 6=-4-2=-6$. Maxraj: $\dfrac{1}{3}+(-4)=-\dfrac{11}{3}$. $-6:\left(-\dfrac{11}{3}\right)=\dfrac{6\cdot 3}{11}=\dfrac{18}{11}=1\dfrac{7}{11}$.

8-MISOL · NATURAL SONLAR YIG'INDISI

Dastlabki n ta natural son yig'indisi $S = \frac{1}{2}n(n+1)$ formula orqali topiladi.

Quvurlar uyum qilib taxlangan: eng pastki qatorda 40 ta, har yuqoriroq qatorda bittadan kam, eng tepada 1 ta. a) Uyumda nechta quvur bor? b) Yuqoridagi 12 qatorni olib ketishdi. Nechta quvur qoldi?

a) Jami: $1 + 2 + \cdots + 40$, $n = 40$: $S = \frac{1}{2} \cdot 40 \cdot 41 = 820$.

b) Olib ketilgani: $1 + 2 + \cdots + 12 = \frac{1}{2} \cdot 12 \cdot 13 = 78$. Qolgani $13 + 14 + \cdots + 40 = 820 - 78 = 742$.

9-MISOL · O'RTACHA TEZLIK

Avtobus 3 soat v km/soat tezlikda, keyin yana 2 soat 60 km/soat tezlikda yurdi. Butun yo'ldagi o'rtacha tezlik qancha?

1. Jami yo'l = tezlik · vaqt: $3v + 2 \cdot 60 = 3v + 120$ km.
2. O'rtacha tezlik = jami yo'l : jami vaqt: $\frac{3v + 120}{5}$ km/soat. $v = 70$ bo'lsa: $\frac{330}{5} = 66$ km/soat.

10-MISOL · O'LCHAMLAR O'ZGARSA

To'g'ri to'rtburchak yuzi $S = ab$. Ikkinchisining eni 3 marta uzun, bo'yi 2 cm qisqa.

1. Formuladagi a o'rniga $3a$, b o'rniga $(b - 2)$ qo'yamiz: $S_2 = 3a(b - 2)$.
2. $a = 4$, $b = 7$: $S_2 = 3 \cdot 4 \cdot 5 = 60$ cm².

11-MISOL · IFODAGA VAZIYAT O'YLAB TOPING

$5x + 2\,000$: masalan, «har biri x so'mlik 5 ta daftar va 2 000 so'mlik paket olindi — jami narx». $\frac{60}{t}$: «60 km yo'l t soatda bosib o'tildi — tezlik».



Manfiy sonni qavssiz qo'ysam, kvadrati manfiy chiqib qoladi! Endi har doim qavs bilan qo'yaman.

DIQQAT!

1. ✗ $x = 5$: $x^2 = 2 \cdot 5 = 10$ ✓ $x^2 = 5 \cdot 5 = 25$
2. ✗ $x = -3$: $x^2 = -3^2 = -9$ ✓ $x^2 = (-3)^2 = 9$
3. ✗ $x = 4$: $3x = 34$ ✓ $3x = 3 \cdot 4 = 12$
4. ✗ $n = 2$: $3n^2 = (3 \cdot 2)^2 = 36$ ✓ $3n^2 = 3 \cdot 2^2 = 12$; $(3n)^2 = 36$ — **boshqa ifoda**
5. ✗ $1y, y3, 100 - 9\pi = 91\pi$ ✓ $y, 3y, 100 - 9\pi$ **shundayligicha qoladi**

Mashqlar

1. Algebraik ifoda bilan yozing.
   a) x va 9 ning yig'indisi b) 15 dan k ni ayirish c) m ning 6 ga ko'paytmasi
   d) a ning b ga bo'linmasi e) t ning kubi f) p ni uchlangani va q ning ayirmasi

2. Qisqa yozing.
   a) $b \cdot 9 \cdot a$ b) $m \cdot m \cdot n \cdot n \cdot n$ c) $x \cdot 1$ d) $n \cdot 4$ e) $2a \cdot 5b \cdot c$
   f) $3t \cdot 4t$ g) $15x : 5y$ h) $m : n \cdot 4$ i) $3u - v : 2$ j) $14k : 7 \cdot n$

3. Bir soatda 60 minut bor. t soat 15 minut necha minut bo'lishini ifoda bilan yozing. $t = 2$ bo'lsa, necha minut?

4. $x = 3$ bo'lsa, hisoblang: a) $4x - 5$ b) $x^2 + x$ c) $2x^2$ d) $(2x)^2$ e) $10 - x^3$

5. $a = -2, b = 5$ bo'lsa, hisoblang: a) $3a + 2b$ b) $ab - b^2$ c) a^2b d) $(a - b)^2$ e) $\dfrac{b - a}{a + b}$ f) $2a^3 - 4$

6. $p = \dfrac{1}{2}, q = -3, r = 4$ bo'lsa, hisoblang: a) $6p - q$ b) pqr c) $r^2 - 8pq$ d) $\dfrac{r + q}{p}$ e) $\dfrac{r}{p} + \dfrac{q}{r}$

7. Formulalarda y ning qiymatini toping.
   a) $y = 2a(b-4),\ a=3,\ b=1$ b) $y = (k+2)^2 - k,\ k=-5$ c) $y = \dfrac{m+n}{m-n},$ $m=3,\ n=-1$
   d) $y = \dfrac{c^2}{2} - 3c,\ c=-4$ e) $y = \dfrac{12}{p^2-1},\ p=-3$ f) $y = h^3 + 2h,\ h=-2$ g) $y = \dfrac{s}{t} + st,\ s=6,\ t=-2$

8. $n = 0, 1, 2, -1$ qiymatlarni qo'yib tekshiring: ifodalar teng kuchlimi?
   a) $n-5$ va $5-n$ b) $n \cdot n$ va n^2 c) $n+n+n+n$ va $4n$ d) n^2+1 va $(n+1)^2$ e) $3n^2$ va $(3n)^2$

9. Sport do'konida futbolka x so'm, kepka y so'm turadi. Murabbiy 4 ta futbolka va 2 ta kepka olib, kassaga 100 000 so'm berdi.
   a) Jami to'lovni va qaytimni x, y orqali yozing. b) $x = 18\,000$, $y = 9\,000$ bo'lsa, qaytim qancha?

10. Velosiped ijarasi: birinchi soati 15 000 so'm, keyingi har soati 10 000 so'm. t soatlik ijara narxini yozing ($t \geq 1$). 4 soat necha so'm?

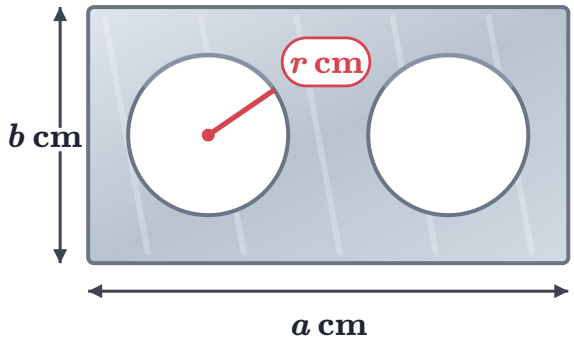
11. Uch javonda kitob bor: 1-javonda k ta, 2-javonda 1-javondagidan 2 marta ko'p, 3-javonda 2-javondagidan 9 ta kam.
    a) Har bir javondagi kitoblar sonini k orqali yozing. b) $k = 15$ bo'lsa, 3-javonda nechta kitob bor?

12. Bo'yi a cm, eni b cm bo'lgan karton varaqning to'rtta burchagidan tomoni c cm bo'lgan kvadratlar qirqib olindi. a) Qolgan qismning yuzini yozing. b) $a = 20$, $b = 12$, $c = 3$ bo'lsa, uni toping.

13. Farengeyt shkalasidagi F haroratni Selsiy shkalasiga o'tkazish formulasi: $C = \dfrac{5}{9}(F - 32)$. Selsiyda ifodalang: a) 50 °F b) 14 °F c) 95 °F

14. $S = \dfrac{1}{2}n(n+1)$ formulasidan foydalanib toping: a) $1 + 2 + \cdots + 30$ b) $26 + 27 + \cdots + 60$ c) $3 + 6 + 9 + \cdots + 90$

15. Bo'yi a cm, eni b cm li metall listdan radiusi r cm bo'lgan ikkita doira qirqib olindi.

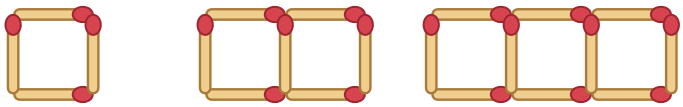


a) Qolgan qismning yuzini a, b, r va π orqali yozing. b) $a = 30$, $b = 16$, $r = 5$ bo'lsa, yuzini π orqali ifodalang.

16. a) Bir xil 12 ta kitob ustma-ust taxlandi, taxlam balandligi h cm. Bitta kitobning qalinligini h orqali yozing.
b) To'g'ri to'rtburchakning yuzi 48 cm², eni x cm. Bo'yini x orqali yozing.

17. Avtobusda a nafar yo'lovchi bor edi. Bekatda b nafari tushdi, chiqqanlar esa tushganlarning 2 baravaridan 3 nafar ko'p edi. a) Avtobusda endi nechta yo'lovchi bor? b) $a = 25$, $b = 6$ bo'lsa-chi?

18. Gugurt cho'plaridan qatorasiga kvadratlar terildi.



1-shakl 2-shakl 3-shakl

Shakl raqami	1	2	3	4	8	n
Cho'plar soni	4	7	10	?	?	?

a) Jadvalni to'ldiring. b) 30 ta kvadrat uchun nechta cho'p kerak? c) 100 ta cho'pdan nechta kvadrat chiqadi?

19. Bog'dagi 20 tup olma daraxtining har biridan o'rtacha m kg, yana 5 tup yosh daraxtning har biridan o'rtacha 12 kg hosil olindi.
a) Jami hosilni m orqali yozing. b) Bir tupga o'rtacha qancha hosil to'g'ri keladi? c) $m = 40$ bo'lsa, buni hisoblang.

20. Uzunligi x m bo'lgan tasmadan 3 ta sovg'a qutisini bog'lash uchun har biriga y m ketdi. Qolganini k nafar bola teng bo'lib oldi.
a) Har bir bolaga necha metr tasma tegdi? b) $x = 12$, $y = 1{,}5$, $k = 5$ bo'lsa-chi?

21. Kuboid hajmi $V = abc$. Ikkinchi kuboidning bo'yi birinchisidan 2 marta uzun, eni 3 cm qisqa, balandligi bir xil.
    a) Ikkinchi kuboid hajmini a, b, c orqali yozing. b) $a = 5, b = 8, c = 4$ bo'lsa, uni toping.

22. Quyidagi ifodalar bilan yoziladigan hayotiy vaziyat o'ylab toping: a) $3x + 2y$ b) $100 - 4k$ c) $\frac{s}{20}$

5-QISM

Tekshiramiz

1. Algebraik ifoda bilan yozing: a) a ning kvadratidan b ni ayirish; b) m va n ning yig'indisini 4 ga bo'lish; c) x ning qarama-qarshisidan y ning kubini ayirish.

2. Qisqa yozing: a) $6 \cdot c \cdot d$ b) $k \cdot k \cdot k \cdot k$ c) $2a \cdot 3a$ d) $p : q \cdot r$

3. $x = -2, y = 3$ bo'lsa, hisoblang: a) $5x + 2y$ b) x^2y c) $(x - y)^2$ d) $\frac{y - x}{x + y}$

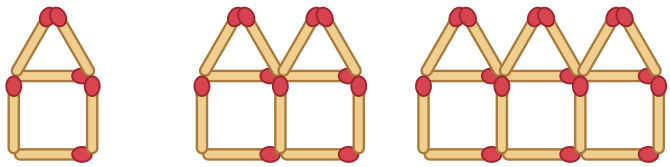
4. $m = \frac{1}{4}$ bo'lsa, hisoblang: a) $8m + 1$ b) $16m^2$

5. $S = \frac{1}{2}n(n + 1)$ formulasi bilan $1 + 2 + \cdots + 24$ ni toping.

6. Mashina bakida 50 litr benzin bor. Har bir kilometrga 0,08 litr sarflanadi. s km yurgandan keyin bakda qancha benzin qolishini yozing. $s = 300$ bo'lsa-chi?

7. n^3 va $3n$ teng kuchlimi? Javobingizni bitta son bilan asoslang.

8. Rasmdagi uychalar qatori uchun cho'plar sonini yozing. 8 ta uycha uchun nechta cho'p kerak?



1-shakl 6 ta cho'p | **2-shakl** 11 ta cho'p | **3-shakl** 16 ta cho'p

9. To'g'ri to'rtburchak gilamning bo'yi a m, eni b m, o'rtasida radiusi r m dumaloq naqsh bor. Naqshdan tashqari qism yuzini yozing va $a = 4, b = 3, r = 1$ da π orqali ifodalang.

10. Aziz: «$x = -5$ bo'lsa, $x^2 = -25$», — dedi. Aziz haqmi?

- [ ] Men jumlani va hayotiy vaziyatni algebraik ifoda bilan yoza olaman.
- [ ] Men ko'paytirish belgisini tushirib, daraja va kasr bilan qisqa yoza olaman; a^2 va $2a$ ni adashtirmayman.
- [ ] Men manfiy son va kasrni qavsda o'rniga qo'yib, ifoda qiymatini topa olaman.
- [ ] Men formulaga qiymatlarni qo'yib hisoblay olaman, π li javobni π bilan yozaman.

Javoblar

4-qism · Mashqlar

1-masala

a) $x + 9$ b) $15 - k$ c) $6m$ d) $\frac{a}{b}$ e) t^3 f) $3p - q$

2-masala

a) $9ab$ b) m^2n^3 c) x d) $4n$ e) $10abc$ f) $12t^2$ g) $\frac{3x}{y}$ h) $\frac{4m}{n}$ i) $3u - \frac{v}{2}$ j) $\frac{14k}{7} \cdot n = 2kn$

3-masala

$60t + 15$ minut; $t = 2$: $60 \cdot 2 + 15 = 135$ minut.

4-masala

a) $4 \cdot 3 - 5 = 7$ b) $9 + 3 = 12$ c) $2 \cdot 9 = 18$ d) $6^2 = 36$ e) $10 - 27 = -17$

5-masala

a) $-6 + 10 = 4$ b) $-10 - 25 = -35$ c) $4 \cdot 5 = 20$ d) $(-7)^2 = 49$ e) $\frac{7}{3} = 2\frac{1}{3}$ f) $2 \cdot (-8) - 4 = -20$

6-masala

a) $3 + 3 = 6$ b) $\frac{1}{2} \cdot (-3) \cdot 4 = -6$ c) $16 - (-12) = 28$ d) $1 : \frac{1}{2} = 2$ e) $8 - \frac{3}{4} = 7\frac{1}{4}$

7-masala

a) $6 \cdot (1 - 4) = -18$ b) $(-3)^2 - (-5) = 9 + 5 = 14$ c) $\frac{2}{4} = \frac{1}{2}$ d) $\frac{16}{2} + 12 = 20$ e) $\frac{12}{8} = 1\frac{1}{2}$ f) $-8 + (-4) = -12$ g) $-3 + (-12) = -15$

8-masala

a) Yo'q ($n = 0$: -5 va 5). b) Ha ($n \cdot n$ ning o'zi n^2). c) Ha. d) Yo'q ($n = 1$: 2 va 4). e) Yo'q ($n = 1$: 3 va 9).

9-masala

a) To'lov $4x + 2y$ so'm, qaytim $100\,000 - (4x + 2y)$ so'm. b) $72\,000 + 18\,000 = 90\,000$; qaytim 10 000 so'm.

10-masala

$15\,000 + 10\,000(t - 1)$ so'm; $t = 4$: $15\,000 + 30\,000 = 45\,000$ so'm.

11-masala

a) k, $2k$, $2k - 9$. b) $2 \cdot 15 - 9 = 21$ ta.

12-masala

a) $ab - 4c^2$ cm². b) $240 - 36 = 204$ cm².

13-masala

a) $\frac{5}{9} \cdot 18 = 10$ °C b) $\frac{5}{9} \cdot (-18) = -10$ °C c) $\frac{5}{9} \cdot 63 = 35$ °C

14-masala

a) $\frac{1}{2} \cdot 30 \cdot 31 = 465$. b) $\frac{1}{2} \cdot 60 \cdot 61 - \frac{1}{2} \cdot 25 \cdot 26 = 1\,830 - 325 = 1\,505$. c) $3 \cdot (1 + 2 + \cdots + 30) = 3 \cdot 465 = 1\,395$.

15-masala

a) $ab - 2\pi r^2$ cm². b) $480 - 50\pi$ cm².

16-masala

a) $\frac{h}{12}$ cm. b) $\frac{48}{x}$ cm.

17-masala

a) $a - b + (2b + 3)$. b) $25 - 6 + 15 = 34$ nafar.

18-masala

a) 4 → 13, 8 → 25, $n \to 3n+1$. b) $3 \cdot 30 + 1 = 91$ ta. c) $3n+1 = 100$, $n = 33$ ta kvadrat.

19-masala

a) $20m + 60$ kg. b) $\dfrac{20m+60}{25}$ kg. c) $\dfrac{860}{25} = 34{,}4$ kg.

20-masala

a) $\dfrac{x - 3y}{k}$ m. b) $\dfrac{12 - 4{,}5}{5} = 1{,}5$ m.

21-masala

a) $V_2 = 2a(b-3)c$. b) $2 \cdot 5 \cdot 5 \cdot 4 = 200$ cm³.

22-masala

Masalan: a) x so'mlik 3 ta non va y so'mlik 2 ta sut — jami narx. b) 100 sahifali kitobdan har kuni 4 sahifadan k kun o'qilgandan keyin qolgan sahifalar. c) s km yo'lni 20 km/soat tezlikda bosib o'tish vaqti.

5-qism · Tekshiramiz

1-masala

a) $a^2 - b$ b) $\dfrac{m+n}{4}$ c) $-x - y^3$

2-masala

a) $6cd$ b) k^4 c) $6a^2$ d) $\dfrac{pr}{q}$

3-masala

a) $-10 + 6 = -4$ b) $4 \cdot 3 = 12$ c) $(-5)^2 = 25$ d) $\dfrac{5}{1} = 5$

4-masala

a) $2 + 1 = 3$ b) $16 \cdot \dfrac{1}{16} = 1$

5-masala

$\frac{1}{2} \cdot 24 \cdot 25 = 300.$

6-masala

$50 - 0{,}08s$ litr; $s = 300$: $50 - 24 = 26$ litr.

7-masala

Yo'q: $n = 1$ da $n^3 = 1, 3n = 3$.

8-masala

Har yangi uychaga 5 ta cho'p qo'shiladi: $5n + 1$. $n = 8$: 41 ta cho'p.

9-masala

$ab - \pi r^2$; $(12 - \pi)$ m².

10-masala

Yo'q: manfiy sonni qavs ichida qo'yish kerak: $(-5)^2 = 25$.